

XIANGYU YIN

PHD CANDIDATE

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PITTSBURGH, PA, USA

RESEARCH INTERESTS

AI in Healthcare, Mobile Sensing, AI Applications, AIoT, Robotics.

EXPERIENCE

2019 - Present

Graduate Student Researcher / Teaching Assistant

Intelligent System Lab, Department of Electrical & Computer Engineering
University of Pittsburgh, Pittsburgh, PA, USA

- Developing vision-based AI applications to ease prosthesis alignment for individuals with disabilities.
- Collaborated with pulmonologists at UPMC Children's Hospital of Pittsburgh and developed smartphone-based acoustic sensing systems and machine learning tools for evaluating pulmonary diseases (COPD, asthma, etc.).
- Provided technical support across multiple projects, focusing on the implementation and deployment of machine learning models (especially LLMs) and enhancing their usability, explainability, and efficiency.

2018 - 2019

Research Assistant

Department of Automation

University of Science and Technology of China (USTC), Hefei, China

- Developed hardware-in-the-loop (HIL) simulation tools for testing UAV flight control system.

EDUCATION

Present

Ph.D. in Electrical & Computer Engineering

University of Pittsburgh, Pittsburgh, PA, USA

Supervised by Dr. Wei Gao

June 2019

B.Eng. in Automation

University of Science and Technology of China (USTC), Hefei, China

Enrolled in the *Talent Program in Information Science and Technology*

Graduated from the *School of the Gifted Young*

PUBLICATIONS

1. **[ICCV'25]** Yin, X., Yang, B., Liu, W., Xue, Q., Alamri, A., Fiedler, G., & Gao, W. (2025). ProGait: A Multi-Purpose Video Dataset and Benchmark for Transfemoral Prosthesis Users. *arXiv preprint arXiv:2507.10223*. <https://doi.org/10.48550/arXiv.2507.10223>
2. **[MobiSys'25]** Wang, H., Yang, B., Yin, X., & Gao, W. (2025). Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection. <https://doi.org/10.48550/arXiv.2504.13938>
3. **[arXiv]** Song, J., Huang, K., Yin, X., Yang, B., & Gao, W. (2024). Achieving Sparse Activation in Small Language Models. *arXiv preprint arXiv:2406.06562*. <https://doi.org/10.48550/arXiv.2406.06562>

4. **[CVPR'25]** Xue, Q., **Yin, X.**, Yang, B., & Gao, W. (2025). Phyt2v: Llm-guided iterative self-refinement for physics-grounded text-to-video generation. In *Proceedings of the Computer Vision and Pattern Recognition Conference* (pp. 18826-18836). <https://doi.org/10.48550/arXiv.2412.00596>
5. **[MobiCom'25]** Huang, K.*, **Yin, X***, Huang, H., & Gao, W. (2025). Modality plug-and-play: Runtime modality adaptation in LLM-driven autonomous mobile systems. In *ACM MobiCom*. https://sites.pitt.edu/~weigao/publications/mobicom25_mnpn.pdf
6. **[MobiCom'24]** Huang, K., **Yin, X.**, Gu, T., & Gao, W. (2024). Perceptual-Centric Image Super-Resolution using Heterogeneous Processors on Mobile Devices. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking* (pp. 1361-1376). <https://doi.org/10.1145/3636534.3690698>
7. **[MobiSys'23]** **Yin, X.**, Huang, K., Forno, E., Chen, W., Huang, H., & Gao, W. (2023). PTEase: Objective Airway Examination for Pulmonary Telemedicine using Commodity Smartphones. In *Proceedings of the 21st Annual International Conference on Mobile Systems, Applications and Services* (pp. 110-123). <https://doi.org/10.1145/3581791.3596854>
8. **[CML-IOT'22/SenSys'22]** **Yin, X.**, Huang, K., Forno, E., Chen, W., Huang, H., & Gao, W. (2022). Out-Clinic Pulmonary Disease Evaluation via Acoustic Sensing and Multi-Task Learning on Commodity Smartphones. In *Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems* (pp. 1182-1188). <https://doi.org/10.1145/3560905.3568437> (**Best Paper Award**)

* Equal contribution

PROFESSIONAL ACTIVITIES

Presentations:

- “Perceptual-Centric Image Super-Resolution using Heterogeneous Processors on Mobile Devices”, The 30th Annual International Conference on Mobile Computing and Networking (MobiCom'24), Washington, D.C., November 2024
- “Smartphone-Based Acoustic Waveform Airway and Respiratory Examination”, ATS 2023 International Conference, Washington, D.C., May 2023 (Poster)
- “Out-Clinic Pulmonary Disease Evaluation via Acoustic Sensing and Multi-Task Learning on Commodity Smartphones”, The Fourth Workshop on Continual and Multimodal Learning for Internet of Things (CML-IOT'22), Co-Located with SenSys 2022, Boston, MA, November 2022
- “Acoustic Waveform Respiratory Evaluation (AWARE)”, i4Kids Symposium, UPMC Children's Hospital of Pittsburgh, Pittsburgh, PA, June 2022
- “Neural Network Memoization for Scalable Edge Inference”, Elijah Meeting, Dept. of Computer Science, Carnegie Mellon University, Pittsburgh, PA, March 2022

Conference Reviewer:

- 2023 IEEE/ACM international conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE'23)
- 2023 IEEE International Conference on Communications (ICC'23) E-health Track
- 2023 IEEE International Conference on Computer Communications (INFOCOM'23)
- The 19th IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS'22)

SKILLS & ABILITIES

Programming Skills: MATLAB, C/C++, Python, Java, JavaScript, HTML, SQL, LaTeX

Software & Tools: MATLAB, PyTorch, Keil, Autodesk Fusion, UltiMaker Cura

Hardware Platforms: Android, STM32, Raspberry Pi, Nvidia Jetson

HONORS & AWARDS

November 2024	Student Travel Grant ACM MobiCom'24
November 2022	Best Paper Award ACM CML-IOT'22
December 2017	Bronze Prize of Scholarship for Outstanding Students in USTC University of Science and Technology of China (USTC)
August 2017	National Second Prize / Provincial First Prize National Undergraduate Electronic Design Contest
December 2016	Seagate Scholarship of USTC University of Science and Technology of China (USTC)